KCPC Workshop 1
Problems

1 Introductory brainteaser

1.1 Left handed people

In a room of 100 people, 99% are left handed.

Question: How many people should leave the room to bring the percentage down to 98%?

2 Induction and Logic Puzzles

2.1 The Locker Problem

There are 100 lockers in a high school with 100 students.

- The first student opens all lockers.
- The second student closes all even-numbered lockers (2, 4, 6, 8...).
- The third student toggles every third locker (closes if open, opens if closed).
- The n-th student toggles every n-th locker ($n, 2n, 3n \dots$).

This continues until all 100 students have had a turn.

Question 1: How many lockers are open at the end of this exercise?
Question 2: What is the general answer for N lockers and N students?

2.2 The Three Wise Men

The King called the three wisest men to his court to decide who would become his new advisor. He placed a hat on each head such that each man could see the other two hats, but not his own. Each hat was either white or blue. The king guaranteed that at least one person was wearing a blue hat. The contest was declared fair to all three men, and they were forbidden from speaking. After sitting in silence for a long time, one man stood up and correctly announced the color of his hat.

Question: What color did he announce, and how did he deduce it?

2.3 Muddy Children

A group of attentive children is told that some of them (at least one) have muddy faces. Each child can see the faces of the others, but cannot tell if his or her own face is muddy. The children are told that those with muddy faces must step forward, but any child with a clean face who steps forward will be punished. At the count of three, every child who believes that his or her face is muddy must step forward simultaneously; any child who signals to another in any way will be punished. If any child with a muddy face has not stepped forward, the process will be repeated. At a given iteration, all muddy children and only them step forward.

Question: What is their thought process and on which turn do they step forward?

3 Invariance and Physics

3.1 Ants on a Ruler

100 ants are dropped simultaneously at arbitrary positions on a straight ruler of fixed length L. Each ant walks at a constant speed v, choosing an initial direction (left or right) at random. Whenever two ants collide, they instantly reverse directions and continue at the same speed v. Ants fall off the ruler as soon as they reach either end.

Question: Determine the time T one must wait to be certain that all ants have fallen off the ruler, regardless of their initial positions and directions.

3.2 The Fly and the Cyclists

Two cyclists, A and B, are exactly 20 miles apart. They pedal toward each other at a constant speed of 10 mph each. At the moment they start, a fly on Cyclist B's wheel takes flight at 15 mph toward Cyclist A. Upon reaching A, it reverses toward B, continuing this back-and-forth shuttle until the cyclists collide.

Question: What is the total distance the fly traveled before the collision?

4 Game Theory

4.1 Simple Stick Game

Consider a game where there is initially a heap of n sticks. Players A and B move alternately, and player A begins. On each move, the player must remove 1, 2, or 3 sticks from the heap. The player who removes the last stick wins the game.

Question: Determine which player has a winning strategy based on the initial value of n.